BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

My goal is to find the highest possible output for eight unknown numerical functions, using only one new evaluation per function each round. Their inputs have between two and eight dimensions, and the formulas are hidden. Every evaluation is therefore an opportunity both to improve the best result and to learn about the search space.

I began with the supplied observations, reviewed the best points and built Gaussian Process surrogate models using NumPy and scikit-learn. The models estimate outputs and uncertainty; Expected Improvement helped suggest promising candidates. I manually checked each query for the correct dimensions and six-decimal formatting before submitting it. After receiving the outputs, I compared them with the best results, added the observations and adapted the next round. The aim is strong observed performance, not a claim that I have found the unknown global optima.

2. **How your strategy has evolved**
   - Describe the key changes in your approach since your early rounds.
   - What influenced these changes – data trends, feedback, model performance or intuition?
   - Identify which principles or heuristics now guide your query decisions.

Early rounds involved wider exploration because I had little evidence about the functions. As the history grew, I shifted toward local refinement around good results, while retaining some exploration when a function stalled or behaved unexpectedly. Feedback from actual evaluations mattered more than simply following a model prediction.

My decisions are now function-specific. For Function 5, high values across all four dimensions repeatedly performed well, so I exploited that region. For Function 4, I made small changes around a productive combination and continued improving. When Function 6 improved in Week 10 but declined in later tests, I reconsidered the direction rather than assuming each local move would help.

My current rules are: compare against the best observed output, exploit repeatably strong regions, explore when gains stall, avoid unnecessary exact-zero inputs, and check all proposed values manually. I also avoid interpreting a single strong result as proof of a global optimum.

3. **Patterns, data and insights**
   - Highlight the most meaningful trends you've observed across your data points or model evaluations.
   - Identify which variables or behaviours seem to influence results the most.
   - Explain how these observations have shaped your understanding of the search process.

The clearest pattern is that different functions reward different search behaviours. Function 5 rose from an initial best of about 1,089 to 8,282 when all four inputs were near 0.995. Repeating this exact point then produced the same result, suggesting a strong region but little new information from duplicates. Function 4 improved from about -4.026 initially to 0.633 by Week 12 through local refinement. Function 7 reached about 2.436, and Function 8 reached about 9.991.

The most influential behaviours appear to be boundary effects and interactions between variables, rather than one universal 'important' feature. Function 5 favours high inputs; Function 4 responds to a specific four-variable combination; Function 6 shows that nearby points can still have noticeably different outputs. I have not formally measured feature importance, so these are observations rather than proven causal effects.

These patterns showed me why sparse, high-dimensional data require cautious interpretation and why each function needs its own search strategy.

4. **Decision-making and iteration**
   - Explain how you balance exploration and exploitation.
   - Describe one or two examples of strategic decisions. What worked, what didn't, and why?
   - Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

I balance exploration and exploitation according to the evidence for each function. Early on, I tested a wider range of points; later I spent more queries near strong observations, because each evaluation is limited and potentially valuable. I still allow controlled changes when performance stops improving or uncertainty remains high.

One successful decision was repeatedly exploring the high-input region for Function 5: the best observed output increased to 8,282. However, submitting the identical all-high point several times gave no further improvement, so repeating it has low information value. Another example is Function 6: a Week 10 query reached -0.112, but nearby Week 11 and 12 queries fell to -0.180 and -0.216. Local refinement is therefore not guaranteed to improve results.

When observations disagree with expectations, I compare them with the full history, revisit stronger points and avoid overreacting to a single result. Gaussian Process uncertainty helps guide exploration, but the actual evaluations remain the final evidence.

5. **Next steps and reflection**
   - Outline your planned actions or experiments to improve the model's performance.
   - Include two or three sentences connecting your work to the broader ML landscape.
   - Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

For the final round, I would prioritise carefully chosen local refinements around the best observed points for Functions 2, 3, 4 and 8. I would reconsider Function 6 using its Week 10 best, and test a genuinely new nearby point for Function 5 rather than repeating an identical query. I would also compare Gaussian Process suggestions with simpler local perturbations and document all final decisions, parameters and outcomes.

This project reflects a wider machine-learning challenge: choosing useful experiments when labels or evaluations are costly. Bayesian optimisation connects predictive modelling with sequential decision-making, while the higher-dimensional functions show the limits of small datasets. The same principles apply to hyperparameter tuning, experimental design and other expensive optimisation problems.

For a non-technical stakeholder, I would show a simple chart of the best result achieved for each function over time, explain which decisions improved performance, and clearly state that the true maximum is still unknown.